

Gram–Schmidt Orthogonalization

Geometric Foundation of Linear Prediction

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February 2026

Introduction: The Prediction Problem

Given a stochastic process

$$\{X_t\}_{t \in \mathbb{Z}},$$

we want to predict the present value X_t using past observations:

$$X_{t-1}, X_{t-2}, \dots, X_{t-p}.$$

We consider linear predictors of the form:

$$\hat{X}_t = \sum_{j=1}^p \phi_j X_{t-j}.$$

Then,

$$X_t = \sum_{j=1}^p \phi_j X_{t-j} + \epsilon_t.$$

The optimal linear predictor minimizes the mean squared error (MSE):

$$\mathbb{E}[(X_t - \hat{X}_t)^2].$$

Thus, prediction becomes a problem of minimizing the distance between X_t and its approximation.

Geometric Interpretation: Prediction = Projection

Consider the Hilbert space:

$$L^2(\Omega) = \{X : \mathbb{E}[X^2] < \infty\}.$$

In this space:

- Covariance acts as an inner product.
- Variance behaves as a squared norm.
- Uncorrelated variables are orthogonal.

By the projection theorem, the best linear predictor is an orthogonal projection:

$$\hat{X}_t = \text{Proj}_{\mathcal{H}_{t-1}}(X_t),$$

where $\mathcal{H}_{t-1} = \text{span}\{X_{t-1}, X_{t-2}, \dots\}$.

$X_t = \text{Projection onto past} + \text{Orthogonal innovation}.$

Hilbert Space Framework

Define the space as the collection of random variables:

$$\mathcal{H} = L^2(\Omega) = \{X : \mathbb{E}[X^2] < \infty\}.$$

We define the inner product:

$$\langle X, Y \rangle = \mathbb{E}[XY].$$

Then:

- $\mathcal{H}$ is a Hilbert space

This guarantees existence and uniqueness of orthogonal projections.

Norm, Distance, and Orthogonality

Norm:

$$\|X\| = \sqrt{\mathbb{E}[X^2]}$$

Distance (Mean-Square Error):

$$\|X - Y\| = \sqrt{\mathbb{E}[(X - Y)^2]}$$

Orthogonality:

$$\langle X, Y \rangle = 0 \iff \mathbb{E}[XY] = 0.$$

If $\mathbb{E}[X] = \mathbb{E}[Y] = 0$ (Centered):

$$\mathbb{E}[XY] = \text{Cov}(X, Y).$$

$$\mathbb{E}[X^2] = \text{Var}(X)$$

Key Idea: Orthogonality in L^2 corresponds to zero covariance.

Subspace Generated by the Past

For a time series $\{X_t\}$, define:

$$\mathcal{H}_{t-1} = \text{span}\{X_{t-1}, X_{t-2}, \dots\}.$$

This contains all predictors of the form:

$$\sum_{j=1}^p a_j X_{t-j}.$$

If the process is weakly stationary:

$$\langle X_t, X_{t-k} \rangle = \gamma(k).$$

The covariance matrix:

$$\Gamma_p = \begin{pmatrix} \gamma(0) & \gamma(1) & \cdots \\ \gamma(1) & \gamma(0) & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

has **Toeplitz structure**.

Conclusion: Time series prediction is geometry inside a Hilbert space.

Orthogonal Projection in Hilbert Space

Projection Theorem

Let $\mathcal{H}$ be a Hilbert space and $S \subset \mathcal{H}$ a closed subspace. For every $X \in \mathcal{H}$, there exists a unique $\hat{X} \in S$ such that

$$\|X - \hat{X}\| = \inf_{Y \in S} \|X - Y\|.$$

Moreover,

$$X - \hat{X} \perp S.$$

Interpretation:

The best approximation of a vector is its orthogonal projection onto the subspace.

Linear Prediction as Projection

For a time series $\{X_t\}$, define:

$$\mathcal{H}_{t-1} = \text{span}\{X_{t-1}, X_{t-2}, \dots\}.$$

The best linear predictor is:

$$\hat{X}_t = \text{Proj}_{\mathcal{H}_{t-1}}(X_t).$$

Prediction error:

$$e_t = X_t - \hat{X}_t.$$

Orthogonality condition:

$$\langle e_t, X_{t-k} \rangle = 0, \quad k \geq 1.$$

Thus, prediction error is orthogonal to the past.

Derivation of Yule–Walker Equations

Assume finite predictor:

$$\hat{X}_t = \sum_{j=1}^p \phi_j X_{t-j}.$$

Orthogonality implies:

$$\mathbb{E} \left[(X_t - \hat{X}_t) X_{t-k} \right] = 0, \quad k = 1, \dots, p.$$

Expanding and using weak stationarity condition:

$$\gamma(k) = \sum_{j=1}^p \phi_j \gamma(k-j).$$

Matrix form:

$$\Gamma_p \phi = \gamma.$$

Key Insight:

Yule–Walker equations arise directly from orthogonality.

Classical Gram–Schmidt Orthogonalization

Motivation

The past observations

$$X_{t-1}, X_{t-2}, \dots$$

are generally correlated.

Correlated vectors:

- Complicate computations
- Hide geometric structure

Goal:

Transform a set of linearly independent vectors

$$v_1, v_2, \dots, v_n$$

into an orthogonal set spanning the same space.

Gram-Schmidt Construction

Step 1:

$$u_1 = v_1.$$

Step 2:

$$u_2 = v_2 - \frac{\langle v_2, u_1 \rangle}{\langle u_1, u_1 \rangle} u_1.$$

General step:

$$u_k = v_k - \sum_{j=1}^{k-1} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j.$$

Result:

$$\langle u_k, u_j \rangle = 0 \quad (j < k).$$

Geometric Interpretation

Each vector decomposes as:

$$v_k = \text{Projection onto previous space} + \text{Orthogonal component.}$$

The orthogonal component represents:

- New direction
- New independent information

Important Insight for Time Series:

$$X_t = \text{Predictable part} + \text{Innovation.}$$

Gram–Schmidt extracts the innovation step by step.

Gram–Schmidt for Time Series

We apply Gram–Schmidt to the sequence:

$$\{X_t, X_{t-1}, X_{t-2}, \dots\}$$

in the Hilbert space $L^2(\Omega)$.

Define the innovation:

$$\varepsilon_t = X_t - \text{Proj}_{\mathcal{H}_{t-1}}(X_t).$$

Then:

$$\varepsilon_t \perp \mathcal{H}_{t-1}.$$

Thus,

$$\mathbb{E}[\varepsilon_t X_s] = 0 \quad \text{for all } s < t.$$

Innovations Representation

Using Gram–Schmidt recursively:

$$X_t = \sum_{j=1}^{\infty} \psi_j \varepsilon_{t-j} + \varepsilon_t.$$

Each observation decomposes into:

- Predictable component (from past innovations)
- New orthogonal innovation

The sequence $\{\varepsilon_t\}$ is orthogonal:

$$\mathbb{E}[\varepsilon_t \varepsilon_s] = 0 \quad (t \neq s).$$

Conclusion

- Linear prediction in time series is an orthogonal projection problem in a Hilbert space.
- The space $L^2(\Omega)$ provides the geometric structure: covariance as inner product and variance as norm.
- The Yule–Walker equations arise directly from orthogonality conditions.
- Gram–Schmidt orthogonalization transforms correlated past observations into orthogonal innovations.

References



Shumway, R. H., & Stoffer, D. S. (2017). *Time Series Analysis and Its Applications: With R Examples* (4th ed.). Springer.



Brockwell, P. J., & Davis, R. A. (2016). *Introduction to Time Series and Forecasting* (3rd ed.). Springer.